

Autonomous navigation of a racing car

AU516 SAA - Projet de cursus

Teacher : Johvany Gustave

CONTEXT

The objective of this project is to implement a robust autonomous navigation approach enabling a ground vehicle to complete several laps on an unknown circuit without colliding with obstacles. Each vehicle is equipped with a LiDAR, a steering mechanism, and an embedded Raspberry Pi 5 running Ubuntu 24.04 and ROS2. The entire project will be carried out using a physical pedagogical platform.

LEARNING OUTCOMES

1. Perform a state-of-the-art review to design an advanced autonomous navigation solution.
2. Apply LiDAR-based perception and processing methods for robotic applications.
3. Develop a complex multi-process ROS2 system integrating perception and control.
4. Optimize the implementation to make the vehicle responsive, stable, and efficient.

INSTRUCTIONS

You will be divided into **6 groups**. Each group must submit on Moodle the following elements (deadline announced later):

1. A PDF report (up to 20 pages) in English including:
    - Introduction
    - Related works on these topics
    - Proposed work with the algorithmic of the developed approach
    - Obtained results
    - Conclusion
2. A video proving the successful work or the latest achievement.
3. All implemented programs, **properly documented**.
4. A structured PowerPoint presentation in English including flowcharts and/or software-generated drawings illustrating your implementation.

During the last session, each group will present its work in English (20 minutes), followed by a 5-minute question period.

Finally, a competition will be organized to evaluate the performance of your approach under real-world conditions.

1 SPECIFICATIONS

Vehicle presentation

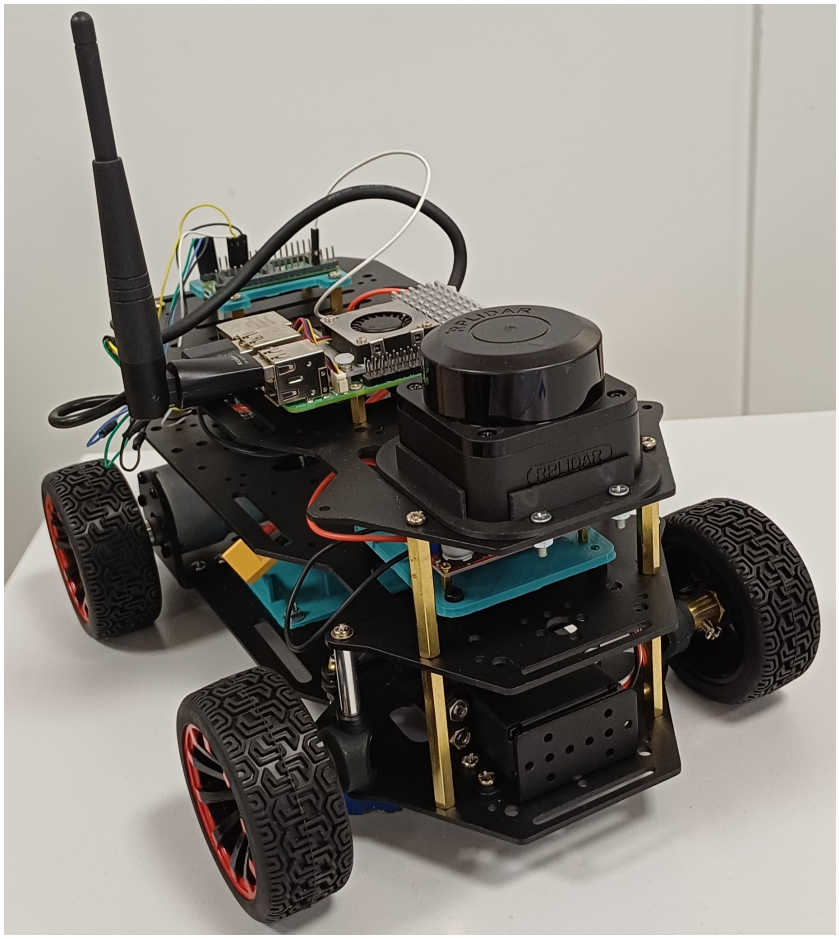


Figure 1: Vehicle presentation

The racing car is composed of:

- Two 12V DC motors with encoders
- A servomotor for steering
- A Raspberry Pi 5 (RPi5) running Ubuntu 24.04 + ROS2
- A Raspberry Pi Pico W handling low-level motor control
- A motor driver (speed control + battery monitoring)
- A LiDAR sensor (RPLIDAR C1) mounted at the front of the vehicle
- A serial communication link between the Pi and the Pico

A detailed description of the vehicle and its wireless programming interface is provided in the document "vehicle_manipulation.pdf" available on Moodle.

Circuit

The final evaluation consists of a race on a predefined circuit delimited by VMC pipes.

The circuit will be circular (ring-like) but its exact configuration will remain unknown until the final race.

Students are therefore asked to anticipate:

- Turns
- Chicanes
- Variable track width

A test track will be available during the project for real-life experimentation.

The race is organized as a time trial. Each vehicle must complete as many laps as possible within a fixed time while:

- staying inside the circuit boundaries
- avoiding collisions with the walls
- maintaining stable and safe motion

Rating

Students will be evaluated according to the following criteria:

- **Technological score**: complexity and relevance of the proposed approach.
- **Project score**: progression and interactions throughout the project.
- **Presentation**: clarity and quality of the final oral presentation.
- **Competition**: performance during the race:
  - Number of completed laps
  - Average lap time
  - Trajectory stability and smoothness
  - Number of collisions or off-track events
  - Overall robustness of the navigation approach

2 GENERAL WORKFLOW AND PROVIDED PROGRAMS

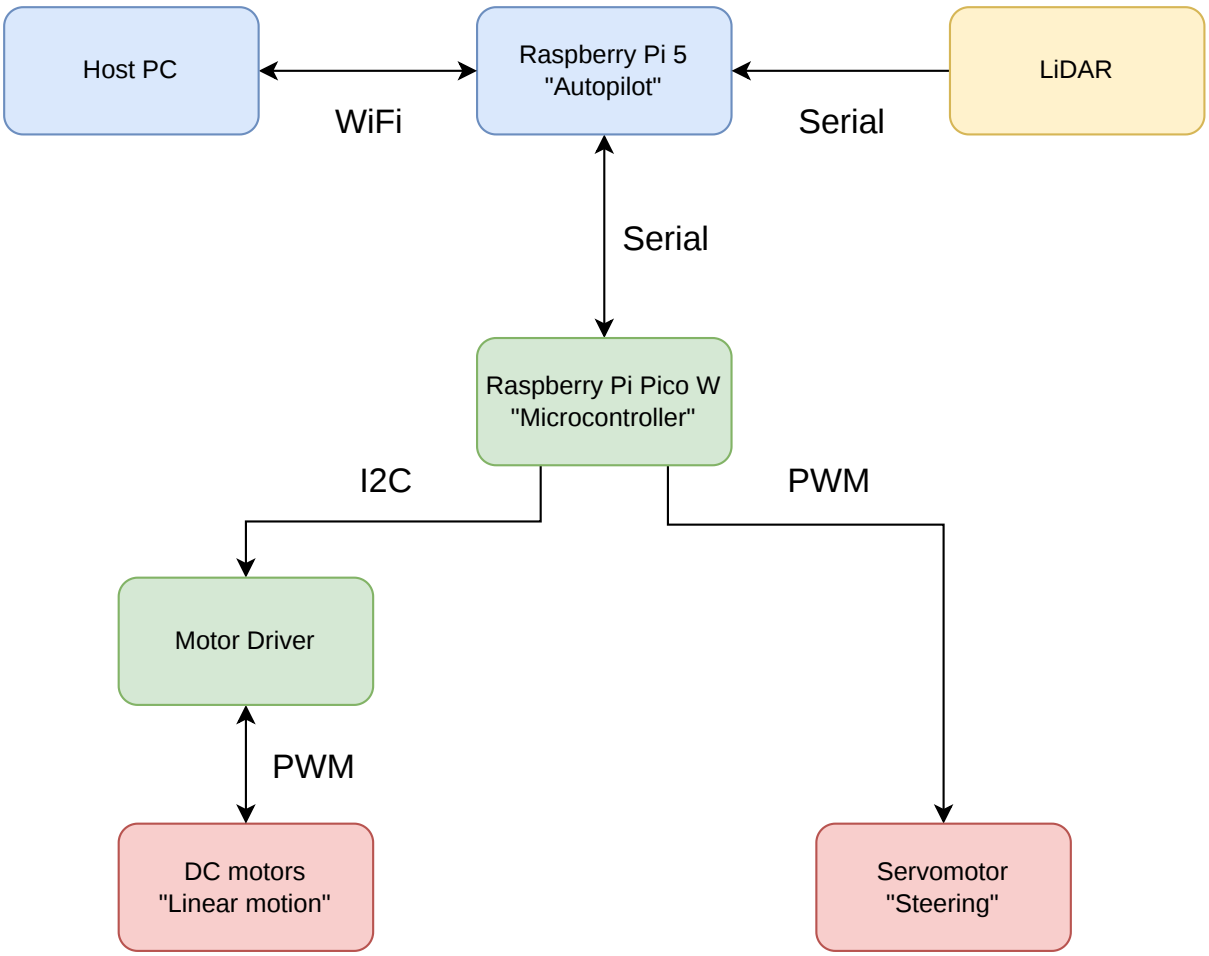


Figure 2: Components interaction

As show in Figure 2, the RPi5 acts as the autopilot, running ROS2 nodes responsible for perception and decision-making. The LiDAR provides 2D range measurements that will be exploited by the RPi5 to determine the target linear velocity of the vehicle and the steering angle. The RPi5 communicates with the Pico W through serial communication, sending high-level commands that the Pico converts into velocity commands sent to the motor driver and PWM command for the servomotor. The RPi5 can communicate with a host PC through WiFi.

LiDAR access (ROS2)

The rplidar_ros package is already installed at:

`~/ros2_ws/src/rplidar_ros`

It publishes LaserScan messages on the `/scan` topic.

A custom ROS2 Python package named `au516_nav` is provided at:

`~/ros2_ws/src/AU516/au516_nav`

It contains a starter node `scan_viewer.py` that subscribes to `/scan` and displays the first ranges of the LiDAR. It serves as a minimal starting point for your perception pipeline.

To start the LiDAR and visualize the first ranges, you must run **two terminals** on the RPi5:

1. Start the LiDAR driver
   In the first terminal:

```
ros2 launch rplidar_ros rplidar_c1_launch.py
```

2. Start the scan viewer node
   In **another terminal**:

```
ros2 run au516_nav scan_viewer
```

Control of the servomotor

PWM is used to control the servomotor involved in the steering mechanism. The servomotor angle is controlled by sending pulses ranging between 1000 and 2000μs.

A pulse of :

- 1500μs puts the front wheels straight.
- 1000μs turns the front wheels completely to the left.
- 2000μs turns the front wheels completely to the right.

The file `"test_servo.py"` **on the Pico W** demonstrates usage.

Control of the motor driver

The file, `"test_driver.py"`, **located in the Pico W storage**, shows how to send velocity commands to the motor driver. It uses the Python module `"motor_driver.py"` implemented by your teacher.

The driver communicates with the Pico W through I²C. To control the speed of the DC motors, use the method `set_speed(speed)`,where `speed` is the target speed for both motors. For each motor, the target speed corresponds to the number of pulses its encoder detects per 10ms. **For your vehicle, the target speed is within the range [-10, 50].**

The motor driver also provides the current battery voltage that supplies the DC motors and the servo-motor. It is expressed in mV. To get it, simply call the method `read_battery_level()`. **This method might return None if there is an error while reading the value.**

Serial communication between the Pico W and the RPi5

Both boards communicate with each other using a TTL-232R-5V-PCB USB-serial interface. This PCB is connected from one side to your Raspberry Pi USB port and from the other side to the **Pico's Serial port 0 (TX: pin 0, RX: pin 1)**. Here is the documentation for the Pico and for the RPi.